

UNIVERSIDAD DE GUADALAJARA

CENTRO UNIVERSITARIO DE LOS VALLES



Maestría en Ingeniería de Software

Course: Administración de la configuración del software

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Baseline v2

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BASELINE v2

StudyGo — Mobile app for gamified homework

PROJECT PROPOSAL

1. THE COMPANY

Name: Blue Mobile Solutions S.A. de C.V.

Location: Guadalajara, Jalisco.

Years in business: 7 years.

Size: 24 employees.

Services: custom mobile app development for Android and iOS.

2. THE CLIENT

Name: Amber High School

School level: high school (general program, 6 semesters)

Type: private school

Location: Guadalajara, Jalisco

Project contact: academic office

3. PROBLEM STATEMENT

The teachers at Amber High School want a mobile app that is easy to use for their students, because all of them have a cell phone but not all of them have a computer.

In the app, each teacher can assign homework for the subject they teach, and the student does it on their phone. The homework is gamified, so the student feels more motivated to do it and to finish all of it.

Teachers can see the scores of each student and of the whole group. This helps them find which topics they need to review again and which ones the students understand better.

The student can see their own progress in each subject, and parents can see the progress and the learning level of their children.

4. PROPOSED SOLUTION

We propose to develop StudyGo, a mobile app for Amber High School.

In the app, the teacher assigns homework for the subject they teach. The student does it on their cell phone. The homework is gamified: the student earns points when they finish it and can see their progress in each subject.

The teacher can see the statistics of their subject. There they see the scores of each student and of the whole group. With this information they find which topics need more review and which ones are better understood.

The parent role is read only. It shows the same information the student sees, but it does not allow doing the homework.

App roles

The app has 3 user roles:

1. Teacher
2. Student
3. Parent

Technology

The client did not say which operating system the teachers, the students and the parents use on their phones. For this reason, the app will be built with Flutter, which is a cross-platform technology.

With Flutter we write one code and from it we build two versions, one for Android and one for iOS. This has two benefits: all users can use the app no matter what phone they have, and the work costs less than making two separate apps.

5. HOW THE APP WORKS

The teacher role allows creating homework, marking the subject and the topic it belongs to. The app has three types of homework: multiple choice, true or false, and filling in a sentence by dragging words. All three are graded by the app.

Gamification means using game elements inside an app that is not a game. It helps the user feel motivated to use it. In this app it is used to motivate the student to do their homework. The student earns points and badges for the homework they finish, goes up a level in each subject, and appears in the ranking of their group.

6. MODULES

The system has 8 modules:

1. Users and login
2. Subjects and groups
3. Homework
4. Points and badges
5. Student progress
6. Teacher reports
7. Parent view
8. Notices

7. SCOPE

The project includes the 8 modules listed in this document, in their version for Android and iOS.

Any other feature that is not on that list is outside this proposal and will be planned for a second version of the system.

8. ESTIMATED TIME

The development takes 5 months. We add 1 more month as extra time for problems, so the project is delivered in 6 months.

The team works with the Scrum method. The work is divided into 10 sprints of 2 weeks each. In every sprint the team agrees to finish one part of the system, and at the end of the sprint the school is shown what is working.

Table 1. Distribution of the work in the sprints.

SPRINTS	WEEKS	WORK
1 and 2	1 to 4	Requirements analysis and screen design
3	5 and 6	Module 1. Users and login
4	7 and 8	Module 2. Subjects and groups
5	9 and 10	Module 3. Homework: create and assign
6	11 and 12	Module 3. Homework: answer the three types
7	13 and 14	Modules 4 and 5. Points and badges, student progress
8	15 and 16	Modules 6 and 7. Teacher reports, parent view
9	17 and 18	Module 8. Notices
10	19 and 20	Final testing and delivery

9. PROJECT TEAM

For this project we assigned 8 people: 3 programmers and 5 people in the other roles.

Table 2. Project team and skills.

ROLE	SKILLS
Project manager	Scrum, planning, Jira
Requirements analyst	Interviews, user stories, UML
UI/UX designer	Figma, Material Design, usability testing
Programmer 1 (technical lead, senior)	Flutter, Dart, Firebase, Git
Programmer 2 (junior)	Flutter, Dart, SQLite, REST services
Programmer 3 (backend, semi-senior)	Node.js, NestJS, PostgreSQL, Docker
QA engineer	Test cases, Postman, Appium
Infrastructure engineer	Docker, cloud services, Linux

10. BUDGET

Table 3. Team salaries.

ROLE	MONTHLY SALARY	MONTHS	TOTAL
Programmer 1 (senior)	\$42,000	6	\$252,000
Programmer 2 (junior)	\$20,000	5	\$100,000
Programmer 3 (semi-senior)	\$30,000	6	\$180,000
Project manager	\$40,000	3	\$120,000
Requirements analyst	\$26,000	2	\$52,000
UI/UX designer	\$24,000	2	\$48,000
QA engineer	\$22,000	3	\$66,000
Infrastructure engineer	\$30,000	2	\$60,000
Salaries subtotal			\$878,000

Table 4. Other expenses of the project.

CONCEPT	AMOUNT
Office rent, the part used by the team ($\$7,000 \times 6$ months)	\$42,000
Internet and utilities ($\$2,000 \times 6$ months)	\$12,000
Test phones	\$25,000
Licenses and cloud services during development	\$29,500
Subtotal	\$108,500

Table 5. Total cost of the project.

CONCEPT	AMOUNT
Salaries subtotal	\$878,000
Other expenses	\$108,500
Company profit (25%)	\$246,625
Total	\$1,233,125 MXN

DESIGN

11. HOW THE MODULES CONNECT

Figure 1 shows how the modules of the system connect to each other.

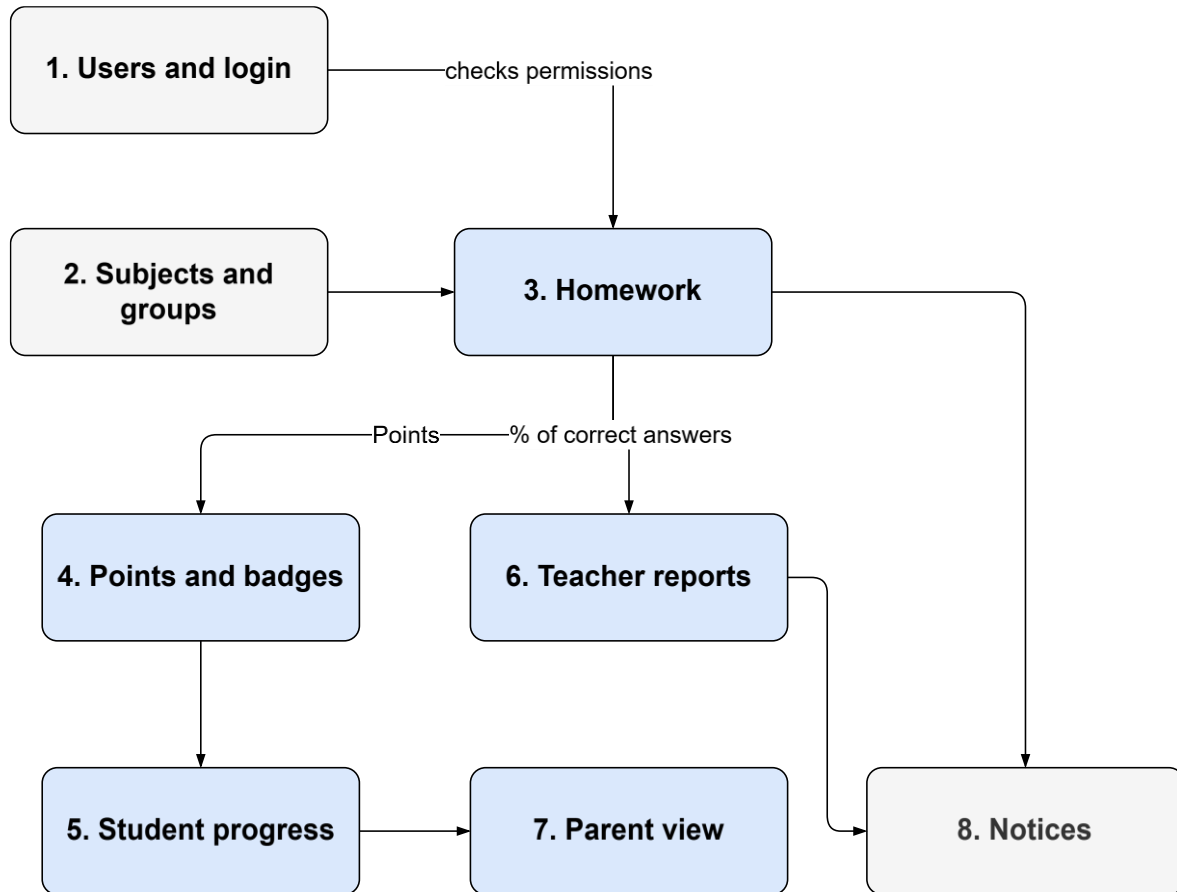


Figure 1. Connection between the modules of the system.

Module 1 is the base. It saves who each user is and what role they have, and the other modules ask it if the user has permission. Module 2 gives module 3 the subjects, the groups and the topics, because every homework belongs to a subject, a topic and a group.

Module 3 is the center of the system. When the student finishes homework, it grades it and saves two numbers: the points and the percentage of correct answers. The points go to module 4, which builds levels, badges and ranking. The percentage of correct answers goes to module 6, which builds the teacher reports.

Module 5 shows that information to the student and module 7 to the parent. Module 8 sends the notifications.

12. DESCRIPTION OF THE MODULES

Module 1. Users and login

Saves the accounts of all the users. Each account has a role and that role sets what the user can do inside the app. It includes the registration, the login and the password recovery.

Table 6. Permissions of each user role.

ROLE	WHAT THEY CAN DO
Teacher	Adds subjects, groups and topics. Creates homework and sees the reports of their groups
Student	Joins their group, answers homework and sees their progress
Parent	Checks the progress of the student. Does not answer homework

Module 2. Subjects and groups

Saves the subjects, the groups and the topics of each subject. It assigns the teachers to the groups where they teach and the students to their group. The topics are needed because every question is marked with a topic, and the reports come from there.

Module 3. Homework

Lets the teacher create homework of the three types, give it a subject, a topic and a due date, and send it to a group. It is also where the student answers the homework and where the system grades it and saves the result.

Table 7. Types of homework.

TYPE OF HOMEWORK	HOW IT IS ANSWERED
Multiple choice	Choose one answer out of four
True or false	Choose one out of two
Fill in the sentence	Drag the words into the blanks (drag and drop)

Module 4. Points and badges

Calculates the points, keeps the level of the student in each subject, gives the badges and builds the ranking.

Table 8. Gamification elements.

ELEMENT	HOW IT WORKS
Points	10 for each correct answer, plus extras for no mistakes and for delivering on time
Levels	Every 500 points the student goes up a level. There are 5 levels per subject
Badges	First homework finished, 100% in one homework, 5 in a row with no mistakes, all the homework of the month
Ranking	The first five places of the group, by points

Module 5. Student progress

Shows the student their progress in each subject: their level and how many points are left for the next one, how much homework they delivered out of the homework assigned, the badges they earned and their place in the ranking.

Module 6. Teacher reports

Shows the teacher the complete list of their group, with the delivered homework and the average of correct answers of each student. These are two different numbers: the delivered homework shows if the student does their work and the average of correct answers shows if they are learning. It also shows the average of correct answers of the group by topic, to know which topics need more review.

Module 7. Parent view

Shows the parent the progress of the student, the same way the student sees it, but read only. It does not allow answering homework.

Module 8. Notices

Sends notifications to the phone when new homework is assigned, when it is close to the due date and when the student gets their grade. It also lets the teacher send notices to their group.

13. ARCHITECTURE

Figure 2 shows the architecture of the system.

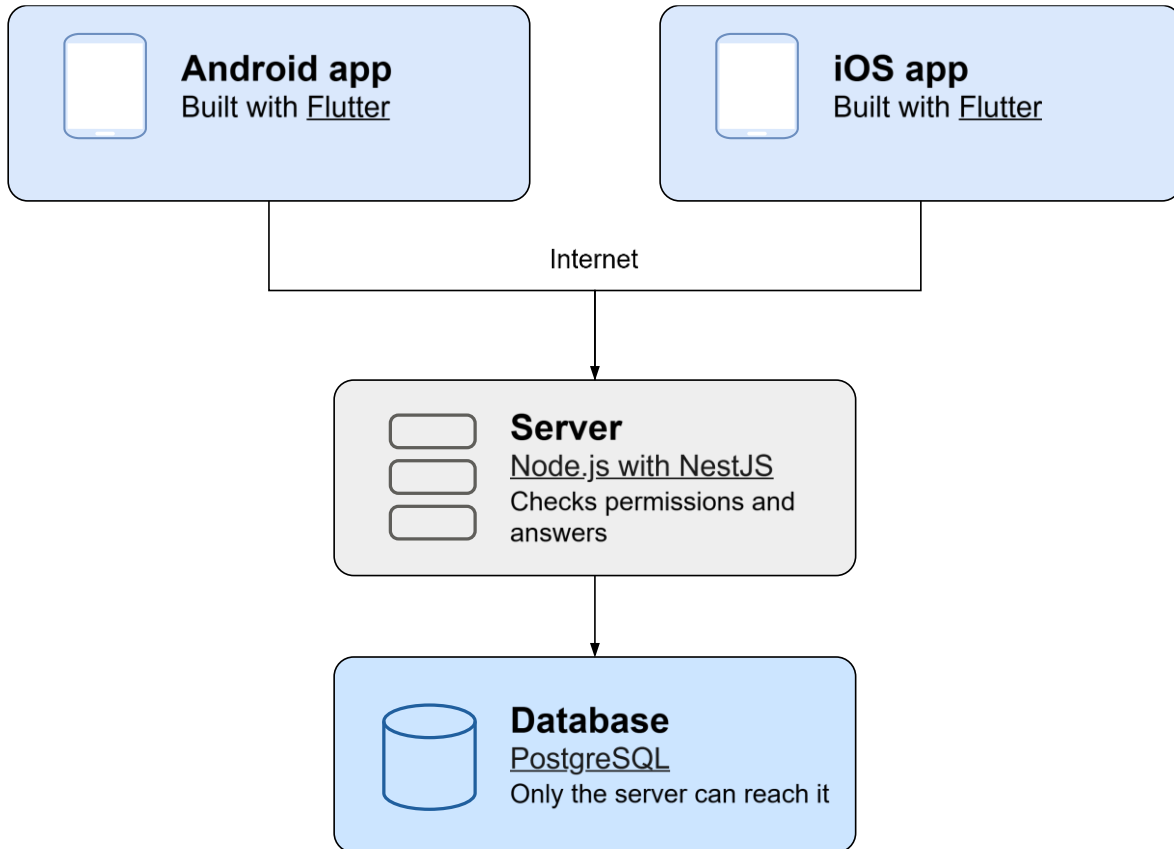


Figure 2. Architecture of the StudyGo system.

The app has three parts. The mobile app is built with Flutter, which is a cross-platform technology: with one code we build the Android and the iOS versions.

The server is built with Node.js and NestJS. The app does not save the information on the phone. Every time the user does something, the app sends the request to the server through the internet. The server checks that the user has permission, does the operation and sends back the answer.

The database is PostgreSQL and it saves all the information. Only the server reaches it, the app never connects directly. This is done for security, so there is only one place where the permissions of each user are checked.

REQUIREMENTS

14. FUNCTIONAL REQUIREMENTS

Table 9. Functional requirements.

ID	NAME	DESCRIPTION
RF-01	User registration	The system lets a person register with a name, an email and a password, choosing the teacher, student or parent role
RF-02	Login	The system lets the user log in with the email and password they registered
RF-03	Password recovery	The system sends an email to reset the password
RF-04	Permissions by role	The system shows each user only the functions that match their role
RF-05	Link parent to student	The parent links their account to the student account by typing a code
RF-06	Create subject	The teacher adds the subjects they teach
RF-07	Create group	The teacher creates a new group or joins one that already exists
RF-08	Join a group	The student joins their group by typing the group code
RF-09	Create topics	The teacher adds the topics of each subject
RF-10	Create homework	The teacher creates homework with a subject, a topic, a type and a due date
RF-11	Multiple choice homework	The teacher writes a question with four answers and marks the correct one
RF-12	True or false homework	The teacher writes a statement and marks if it is true or false
RF-13	Fill in the sentence homework	The teacher writes a sentence with one to three blanks, marks the correct words and adds extra words
RF-14	Assign homework	The teacher sends the homework to the group they choose
RF-15	Answer homework	The student answers the homework on their phone before the due date
RF-16	Automatic grading	The system grades the homework and saves the points and the percentage of correct answers
RF-17	Second try	The student can answer the homework one more time to improve their grade

ID	NAME	DESCRIPTION
RF-18	Show correct answers	The system shows the correct answers when the due date closes
RF-19	Calculate points	The system calculates the points from the correct answers, the mistakes and the delivery time
RF-20	Level up	The system moves the student to the next level when they reach the points needed in a subject
RF-21	Give badges	The system gives a badge when the student reaches an achievement
RF-22	Build ranking	The system sorts the students of the group by points and shows the first five places
RF-23	See progress by subject	The student sees their level, their points and how many points are left for the next level
RF-24	See pending homework	The student sees the homework they have left and the one they already sent
RF-25	See badges and place	The student sees the badges they earned and their place in the ranking
RF-26	See group list	The teacher sees all the students of their group with their delivered homework and their average of correct answers
RF-27	Report by topic	The teacher sees the average of correct answers of the group in each topic of the subject
RF-28	See student detail	The teacher opens the detail of a student and sees their homework and their grades
RF-29	Export report	The teacher downloads the report of their group in a file
RF-30	Check student progress	The parent sees the progress, the points and the homework of the student they are linked to
RF-31	Read only access	The parent role does not allow answering or changing the homework
RF-32	Automatic notifications	The system tells the student when homework is assigned, when it is close to the due date and when they get their grade
RF-33	Teacher notices	The teacher sends a written notice to all the students of their group

15. NON-FUNCTIONAL REQUIREMENTS

Table 10. Non-functional requirements.

ID	NAME	DESCRIPTION
RNF-01	Cross-platform	The app works on Android and iOS with one code, built with Flutter
RNF-02	Response time	The system answers in less than 3 seconds with a normal connection
RNF-03	Password security	Passwords are saved encrypted, nobody can see them
RNF-04	Database access	Only the server reaches the database, never the app
RNF-05	Usability	The screens are simple, so a student can use them without training
RNF-06	Availability	The system is available during and outside school hours
RNF-07	Backups	The information is backed up automatically every day
RNF-08	Language	The app is in Spanish